Scientific Functional Ambiguity in Inverse Problems

Preliminary Mathematical Investigation

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Abstract

Inverse-problem reconstructions are commonly evaluated through errors in the reconstructed state. However, reconstruction accuracy does not necessarily determine the accuracy of a downstream scientific quantity. This preliminary investigation studies the ambiguity of scientific functionals over sets of states that are indistinguishable under the forward operator. Particular attention is given to centroid and, more generally, linear-fractional functionals. The analysis identifies how nullspace geometry, positivity, mass variation, and denominator behaviour affect scientific ambiguity. It also motivates a more specific question for subsequent work: when can local functional sensitivity fail to represent finite ambiguity between observationally indistinguishable states?

1 Motivation

Consider an inverse problem

$$y = Ax,$$

where x is the unknown state, A is a forward operator, and y is the observation. If A is non-injective, different states can produce exactly the same observation. Consequently, reconstructing x is not the only relevant question. In many scientific applications the quantity ultimately used for interpretation is a functional

$$F(x),$$

rather than x itself.

This motivates the question:

> How uncertain can a scientific quantity remain among states that are indistinguishable from the observations?

For an admissible set $\mathcal{X}$, define the feasible fibre

$$\mathcal{F}_y = \{x \in \mathcal{X} : Ax = y\}.$$

An observation-conditioned measure of scientific ambiguity is then

$$\Delta_F(y) = \sup_{x,x' \in \mathcal{F}_y} |F(x) - F(x')|.$$

Thus $\Delta_F(y)$ measures ambiguity in the scientific quantity itself, rather than ambiguity in the reconstructed state.

2 Centroid Functionals

The initial functional studied was the centroid

$$C(x) = \frac{v^\top x}{\mathbf{1}^\top x},$$

where v contains the coordinate values associated with the components of x.

Let $e \in \ker A$. Then

$$A(x + e) = Ax,$$

so x and $x + e$ are observationally indistinguishable whenever both are admissible.

Direct calculation gives the exact finite perturbation identity

$$C(x + e) - C(x) = \frac{(v - C(x)\mathbf{1})^\top e}{\mathbf{1}^\top (x + e)}.$$

This separates two mechanisms affecting centroid ambiguity: the orientation of an invisible perturbation relative to $v - C(x)\mathbf{1}$, and the variation of total mass $\mathbf{1}^\top(x + e)$ along the feasible fibre.

3 Structural Findings

Several structural properties were investigated.

First, the dimension of $\ker A$ alone does not determine scientific sensitivity. Explicit operators with equal nullity were constructed whose nullspaces produced substantially different changes in the same scientific functional. The orientation of the invisible directions relative to the functional is therefore important, not merely their number.

Second, if total mass is constant along a fibre, so that

$$\mathbf{1}^\top e = 0 \qquad \text{for every } e \in \ker A,$$

then the denominator of the centroid is constant. On that fibre,

$$C(x) = \frac{1}{M} v^\top x,$$

and the problem reduces to recovery of a linear functional. In finite dimensions, mass preservation along every nullspace direction is equivalent to

$$\mathbf{1} \in \text{range}(A^\top).$$

Hence the genuinely nonlinear behaviour of the centroid is associated with mass-changing invisible directions.

Positivity constraints were also found to be important. In simple examples, an unconstrained feasible fibre can approach a zero denominator and produce unbounded centroid ambiguity, whereas imposing

$$x \geq 0$$

can restrict the admissible part of the fibre and make the centroid range bounded. Positivity does not automatically reduce functional ambiguity, however: if the functional is already invariant along the relevant nullspace directions, shrinking the feasible geometry need not change its range.

4 Geometry of Finite Perturbations

Along a one-dimensional feasible direction, a ratio functional takes the form

$$f(t) = \frac{a + bt}{c + dt}.$$

Its derivative is

$$f'(t) = \frac{bc - da}{(c + dt)^2}.$$

Away from a zero denominator, the function is therefore either strictly monotone or constant.

The same structure extends to multidimensional feasible fibres. If

$$x(z) = x + Ez,$$

where the columns of E span feasible directions, then a linear-fractional functional has the form

$$F(z) = \frac{a + b^\top z}{c + d^\top z}.$$

For a positive denominator,

$$\nabla F(z) = \frac{(c + d^\top z)b - (a + b^\top z)d}{(c + d^\top z)^2}.$$

An interior stationary point can occur only in the degenerate case where the numerator and denominator are proportional, in which case the functional is constant along the fibre. Thus a nonconstant linear-fractional functional has no genuine interior stationary point on a region where its denominator remains positive. On a compact admissible fibre, extrema must therefore occur on its boundary.

5 General Ratio Functionals

The centroid analysis extends naturally to

$$F(x) = \frac{p^\top x}{q^\top x}, \qquad q^\top x > 0.$$

For $e \in \ker A$, the exact finite difference is

$$F(x + e) - F(x) = \frac{(p - F(x)q)^\top e}{q^\top (x + e)}.$$

Consequently, F is constant along the feasible fibre through x if

$$p - F(x)q \perp \ker A.$$

Equivalently,

$$p - F(x)q \in (\ker A)^\perp = \operatorname{range}(A^\top).$$

This gives a geometric interpretation of functional identifiability: the relevant functional direction must contain no component along the observationally invisible directions.

6 Local Sensitivity Versus Finite Ambiguity

The preceding analysis suggests comparing finite ambiguity with local sensitivity. Define the local fibre sensitivity

$$L_F(x) = \sup_{\substack{e \in \ker A \\ \|e\| \le 1}} |DF_x[e]|.$$

For the ratio functional,

$$DF_x[e] = \frac{(p - F(x)q)^\top e}{q^\top x}.$$

Comparing this with the exact finite change gives, whenever $DF_x[e] \neq 0$,

$$\frac{F(x+e) - F(x)}{DF_x[e]} = \frac{q^\top x}{q^\top (x+e)}.$$

This identity isolates the source of disagreement between the local and finite descriptions. If

$$q^\top e < 0,$$

the denominator decreases along the invisible direction and the magnitude of the finite change exceeds the corresponding first-order prediction. If $q^\top e = 0$, the two agree exactly, while increasing the denominator reduces the finite change relative to the local prediction.

In the absence of further constraints, the ratio

$$\frac{q^\top x}{q^\top (x+e)}$$

can become arbitrarily large as $q^\top (x+e)$ approaches zero from above. This motivates studying whether physically meaningful constraints prevent such behaviour.

7 Scope of the Current Results

The present results should be regarded as a preliminary mathematical investigation rather than a completed novelty claim. Several components, including linear-fractional optimisation and nullspace/range-space relationships, are closely related to established mathematical theory.

Their role here has been to identify which aspects of the original problem are structural and to narrow the research question. In particular, the comparison between local scientific sensitivity and finite ambiguity over an observation-conditioned feasible fibre appears to provide a more specific direction for continued investigation.

8 Future Work

The immediate next step is to investigate whether physical constraints and the observed data provide quantitative protection against the denominator mechanism identified above.

In particular, the next problem is to study feasible sets satisfying

$$x \ge 0, \qquad Ax = y, \qquad y \neq 0,$$

and determine conditions under which these assumptions imply a data-dependent positive lower bound

$$q^\top x \ge m(y) > 0 \qquad \text{for all } x \in \mathcal{F}_y.$$

Such a bound would potentially give quantitative control of

$$\frac{q^\top x}{q^\top (x+e)}$$

and therefore of the discrepancy between local sensitivity and finite scientific ambiguity.

The planned investigation is to:

1. derive conditions under which local fibre sensitivity controls finite functional ambiguity;
2. construct counterexamples showing when a single local sensitivity measure can substantially underestimate finite ambiguity;
3. determine whether resulting bounds are sharp and which physical assumptions are necessary;
4. study how positivity, the forward operator, and the particular observation y jointly affect functional identifiability; and
5. subsequently investigate whether deterministic functional ambiguity can be related to posterior uncertainty, distinguishing information supplied by the observations from ambiguity resolved primarily through prior assumptions.

The longer-term objective is therefore not simply to measure reconstruction error, but to determine when the available observations genuinely constrain the scientific quantities that are ultimately inferred from a reconstruction.

9 Conclusion

This preliminary work has developed an observation-conditioned framework for studying scientific functionals in non-injective inverse problems. Analysis of centroid and general ratio functionals shows that scientific ambiguity depends strongly on the orientation of invisible directions and, for nonlinear ratio quantities, on denominator variation along the feasible fibre.

The current work has therefore led to a more focused research question:

When does local scientific sensitivity reliably represent finite same-data functional ambiguity?

Answering this question under physically meaningful constraints forms the next stage of the project.